🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

**UNIT 3 · STANDARD 6.AT.1**

# Understand Ratios

Lesson 3-1 · Teacher Copy

**Name:** Type your name **Date:** Today's date **Class:** Class period

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**CHEF ACADEMY MISSION**

## The Signature Recipe

You just earned your apron at Chef Academy. Head Chef Reyes is trusting you with her signature cookie recipe — 3 cups of flour for every 2 cups of sugar. Before service begins, you must read every recipe like a pro by mastering ratios, or the whole kitchen falls behind.

## My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

**Content Objective: I can write and describe a ratio that compares two quantities.**
**Language Objective: I can explain a comparison using the words ratio, part-to-part, part-to-whole, and colon notation.**

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Understand Ratios | Ratio | Comparison | Part-to-part | Part-to-whole
Colon notation | Tape diagram | Double number line

👆 *Tap any word to see what it means and a picture.*

1. A comparison of two quantities by division is a ______.

2. A comparison of two quantities, like 3 to 5, is a ______.

3. Showing how two amounts relate to each other is a ______.

4. A ratio that compares one part of a group to another part is a ______ ratio.

5. A ratio that compares one part of a group to the total is a ______ ratio.

6 Writing a ratio with a colon, like 3:5, is using ______ .

7 A bar cut into equal parts that shows how a ratio breaks apart is a

______ ______ .

8 Two lines whose matching ticks pair the two amounts in a ratio form a

______ ______

______ .

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**The school cafeteria is planning a new lunch menu. They want to offer 3 main dishes for every 2 side dishes. If they have 10 side dishes, the cafeteria manager needs to figure out how many main dishes to prepare?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Understand Ratios** — Use a ratio to compare two quantities by division.
  *Comprender razones — Usar una razón para comparar dos cantidades mediante la división.*
- [ ] **Ratio** — A way to compare two amounts, like 3 to 2.
  *Razón — Una manera de comparar dos cantidades, como 3 a 2.*
- [ ] **Comparison** — Looking at two or more amounts to see how they are related.
  *Comparación — Mirar dos o más cantidades para ver cómo se relacionan.*
- [ ] **Part-to-part** — A ratio comparing one part of a group to another part.
  *Parte a parte — Una razón que compara una parte de un grupo con otra parte.*
- [ ] **Part-to-whole** — A ratio comparing one part to the whole group.
  *Parte a todo — Una razón que compara una parte con todo el grupo.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.
*Di tu idea a un compañero o en voz baja antes de escribir.*

**For every ____ main dishes the cafeteria serves ____ side dishes, so with 10 side dishes they need ____ main dishes.**

*Di tu respuesta primero: Mi respuesta es ____ porque ____.*

## 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** The recipe uses 3 cups of apple juice for every 2 cups of sparkling water.

**Evidence:** A strong answer names that apple juice and sparkling water are being compared and states the relationship as 3 to 2 (or 'for every 3 apple juice, 2 sparkling water'), not just that there is more apple juice.

**Reasoning:** This evidence connects the definition of understand ratios to the mathematical claim.

### Start

**Most language support**

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **For every ____ main dishes the cafeteria serves ____ side dishes, so with 10 side dishes they need ____ main dishes.**
- **My answer is ____.**
  *Mi respuesta es ____.*

### Build

**Some language support**

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is ____.**
  *Mi afirmación es ____.*
- **I know because ____.**
  *Lo sé porque ____.*
- **So ____.**
  *Entonces ____.*

### Explain

**Light language support**

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is ____.**
  *Mi afirmación es ____.*
- **My math evidence is ____.**
  *Mi evidencia matemática es ____.*
- **This proves ____ because ____.**
  *Esto demuestra ____ porque ____.*

## 4. Check Your Explanation

- [ ] I answered the question.
  *Respondí la pregunta.*
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  *Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- [ ] I explained how the evidence proves my answer.
  *Expliqué cómo la evidencia demuestra mi respuesta.*
- [ ] I used at least two math words.
  *Usé por lo menos dos palabras matemáticas.*
- [ ] I reread complete sentences and punctuation.
  *Volví a leer mis oraciones completas y la puntuación.*

## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

## Answer Key & Teacher Guide

1. **Notes 1:** Understand Ratios
2. **Notes 2:** Ratio
3. **Notes 3:** Comparison
4. **Notes 4:** Part-to-part
5. **Notes 5:** Part-to-whole
6. **Notes 6:** Colon notation
7. **Notes 7:** Tape diagram
8. **Notes 8:** Double number line
9. **Watch for this mistake:** *A common mistake in Understand Ratios is writing the ratio in the wrong order: when asked for the ratio of apple juice to sparkling water in a recipe that uses 3 cups of apple juice for every 2 cups of sparkling water, some students write 2:3 (sparkling water to apple juice) because they list the ingredients in whatever order comes to mind rather than the order the question asks for. The two ratios describe different drinks — 3:2 means 3 cups of juice for every 2 cups of water, while 2:3 would mean only 2 cups of juice for every 3 cups of water. Always match the order of your ratio to the order named in the question: "apple juice to sparkling water" means the apple juice number comes first, giving 3:2.*
10. **Try It 1:** A. 5:12 — *The whole class is 5 + 7 = 12 students. Part-to-whole compares one part to the total, so chefs to whole class is 5:12.*
11. **Try It 2:** A. 9:4 — *"For every 9 honey to 4 lemon" is a part-to-part comparison. In colon notation, honey to lemon is 9:4.*